4. Low Risk Anomaly

The low-risk anomaly is a combination of three effects, with the third a consequence of the first two:[22]

1. Volatility is negatively related to future returns;
2. Realized beta is negatively related to future returns; and
3. Minimum variance portfolios do better than the market.

The risk anomaly is that risk—measured by market beta or volatility—is negatively related to returns. Robin Greenwood, a professor at Harvard Business School and my fellow adviser to Martingale Asset Management, said in 2010, "We keep regurgitating the data to find yet one more variation of the size, value, or momentum anomaly, when the Mother of all inefficiencies may be standing right in front of us—the risk anomaly."

4.1 HISTORY

The negative relation between risk (at least measured by market beta and volatility) and returns has a long history. The first studies showing a negative relation appeared in the late 1960s and early 1970s.[23] Friend and Blume (1970) examined stock portfolio returns in the period 1960–1968 with CAPM beta and volatility risk measures. They concluded (my italics):

> The results are striking. In all cases risk-adjusted performance is dependent on risk. *The relationship is inverse and highly significant*.

Haugen and Heins (1975) use data from 1926 to 1971 and also investigate the relation between beta and volatility risk measures and returns. They report (my italics):

> The results of our empirical effort do not support the conventional hypothesis that risk—systematic or otherwise—generates a special reward. Indeed, our results indicate that, over the long run, *stock portfolios with lesser variance in monthly returns have experienced greater average returns than "riskier" counterparts*.

[22] Some references for the third are Haugen and Baker (1991), Jagannathan and Ma (2003), and Clarke, de Silva, and Thorley (2006). I cover references for the others below.

[23] In addition to the papers in the main text, also see Pratt (1971), Sodolfsky and Miller (1969), and Black (1972).